TexCycle AI: An AI-Powered Textile Waste Marketplace and Circular Economy Intelligence Platform for Sri Lanka

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Abstract—The textile and garment industry directly employs around 350,000 people and generates about USD 5.30 billion in apparel export earnings, making it a significant contributor to Sri Lanka's export economy. [1, 2]. However, waste from the manufacture, processing, distribution, and consumption of textiles poses problems for the environment and the economy. [3, 4]. The AI-powered textile waste marketplace and circular economy intelligence platform TexCycle AI, created specifically for Sri Lanka, is presented in this article. Fabric Intelligence, Sustainability Intelligence, Commercial Intelligence, and Marketplace Predictive Analytics are the four complementary intelligence components that the platform combines into a single process. Uploaded textile photos are transformed into structured properties by Fabric Intelligence, such as fabric type, classification confidence, apparent fault indications, health score, repairability, and remaining lifespan. Sustainability Intelligence converts textile attributes into environmental metrics including energy, water, and CO2 reductions, landfill diversion, and sustainability grades. [5, 6]. Commercial intelligence maintains lifecycle traceability while ranking possible buyer categories and estimating market prices. [7, 8]. Marketplace Predictive Analytics predicts demand, market trends, listing success, and anticipated sales time. React/Vite/Tailwind CSS frontend, Node.js/Express.js backend, MongoDB database, and Python/FastAPI machine-learning services are used in the prototype's implementation. A 100,000-record synthetic marketplace dataset with 36 variables is part of the present prototype. The findings of the experiment show 89.68% test accuracy for fabric-type classification, 90.87% accuracy for defect identification, and price-prediction MAE of 12.24, RMSE of 15.40, and R2 of 0.9590. Demand accuracy was 99.54%, listing success accuracy was 83.65%, market trend accuracy was 98.72%, and buyer matching accuracy was 83.56% in marketplace trials. These findings show technological viability at the prototype level, but they do not prove real-world Sri Lankan market accuracy. The integrated design, clear approach, and requirement for additional validation using actual industry and transaction data are highlighted in the study. [9, 10].

Index Terms: buyer suggestion, pricing prediction, predictive analytics, artificial intelligence, textile waste, circular economy, machine learning, computer vision, textile marketplace, sustainability intelligence, and Sri Lanka.

I. INTRODUCTION

a. Background and Motivation

One of Sri Lanka's most important industrial sectors is the textile and apparel industry. With an estimated USD 5.30 billion in garment export earnings for 2025 and almost 350,000 direct employees, the Sri Lanka Export Development Board recognizes apparel and textiles as a significant contribution to the country's economy. [1, 2]. Due to

the industry's size, there are many opportunities to recover value from textile waste streams and production waste. [4].

In addition to being a threat to the environment, textile waste offers a chance to recover value from resources. Fibres and fabric structures that can be recycled, repaired, upcycled, or reused can still be found in materials that are thrown away, burnt, or dumped in landfills. [3, 11]. Extending material life and decreasing reliance on virgin resources are key components of circular economy research. [3, 11]. However, in reality, a textile waste batch's worth is determined by several variables, such as the kind of material, its condition, its amount, its environmental value, its market price, and its prospective customers. [5, 9].

Textile categorization, sorting, and flaw detection are supported by artificial intelligence and computer vision. [12, 13], and [14]. But identifying a cloth is just one aspect of circular decision-making. Understanding material condition, environmental value, market value, possible purchasers, anticipated demand, and lifetime traceability is also necessary for a complete circular marketplace. [7, 15].

TexCycle AI treats textile waste as a digital circular inventory to meet this demand. The suggested platform combines digital traceability, marketplace forecasting, commercial decision assistance, sustainability analytics, and AI-based material evaluation into a single integrated process.

b. Problem Statement

The way that textile waste is now handled may require some unrelated tasks:

- Textile identification by hand [16]
- Subjective assessment of quality [17]
- Restricted standardization of conditions [17]
- Uncertain pricing methods [4]
- Buyer discovery by hand [14]
- Environmental data that is fragmented [5, 6]
- Limited forecasting of market behavior [9]
- Traceability of disconnected transactions [7]

Determining the best circular pathways for textile materials while still attaining commercial value is challenging due to these fragmentation challenges. [4, 9]. Thus, the research problem is stated as follows:

How can a unified Sri Lankan textile-waste marketplace that enhances material evaluation, environmental decision-making, commercial valuation, buyer discovery, and marketplace prediction be created using artificial intelligence, sustainability analytics, and predictive marketplace technologies?

c. Aim and Objectives

The goal of this research is to create, implement, and assess an AI-powered marketplace for Sri Lankan textile waste that analyses textile materials, estimates environmental value, forecasts commercial prospects, recommends buyers, and predicts marketplace outcomes while maintaining traceability throughout the textile lifecycle.

The goals are:

- Identify textile and fabric kinds automatically using provided photos and confidence data [12, 13].
- Identify obvious flaws and get condition-related metrics, such as repairability and health score [17].
- Use LCA-informed elements and sustainability indicators to estimate environmental value [5, 6].
- Create green and sustainability scores to help make judgements about circular activity [11, 6].
- Estimate the market worth of textiles and suggest appropriate buyer categories [4, 14].
- Use listing, transaction, delivery, and impact records to preserve traceability from AI analysis [7, 8].
- Forecast market trends, sales time, listing success, and demand [9].
- Use the proper classification or regression criteria to assess each AI component [18].
- Examine the platform's suitability in the context of Sri Lanka's textile sector [1, 19, 20].

d. Research Contributions

This study's principal contributions are:

- **C1:** Integrated Textile Intelligence: This system combines condition evaluation, fabric categorization, and defect assessment into a single component [12, 7], and [17].
- **C2:** Sustainability Intelligence: Decisions about the textile trade directly consider environmental indicators and circular action suggestions [5, 20].
- **C3**: Commercial Intelligence: The listing process incorporates AI-based buyer recommendations and price prediction [4, 14].

- **C4:** Marketplace Predictive Intelligence: This software predicts market trends, sales time, listing success, and demand [9].
- **C5:** Sri Lankan Localization: The method takes into account customer classifications, currencies, local districts, provinces, and garment industry standards [1].

The four research components function within a single marketplace process in C6: Complete Circular Marketplace Workflow. [3, 4, 7]:

Textile → AI Analysis → Sustainability Assessment → Price/Buyer Intelligence → Marketplace Prediction → Listing → Transaction → Traceability

I. RELATED WORK

a. Textile Waste and Circular Economy

Strategies for the circular economy seek to prolong the useful life of commodities and reduce dependency on virgin resources. [3, 11]. A recent systematic evaluation of fashion and textile waste management highlighted reuse, recycling, and circular economy transition as significant research concerns, along with supply-chain, stakeholder, technological, and economic hurdles. [4].

Environmental assessment has traditionally placed a strong emphasis on Life Cycle Assessment (LCA), but research on textile circularity shows that the availability of high-quality source data is still a major research shortage. [5]. The predominance of post-consumer textiles and European locations in the literature, according to a thorough 2026 assessment, highlights the need for better primary data and regional-scale research. [9, 10]. This result validates TexCycle AI's emphasis on collecting Sri Lankan textile data instead of depending solely on international sources. [1, 19].

b. AI-Based Textile Classification

Textile categorization is becoming more and more dependent on computer vision and machine learning. [12, 21] Convolutional neural networks and near-infrared spectroscopy are capable of autonomously classifying textile fibers. [13]. AI-based textile sorting using computer vision, machine learning, and spectroscopy has been investigated in more recent works [14, 22].

Nevertheless, blended fiber identification is still difficult, and publicly available databases remain a major obstacle. [14, 23]. Automated textile categorization continues to progress, as evidenced by recent studies comparing CNNs, Vision Transformers, and hybrid architectures for textile fiber identification [18, 24]. This foundation is expanded by TexCycle AI from basic material categorization to condition-aware marketplace intelligence. [12, 17].

c. Defect Detection and Condition Assessment

It takes more than just material identification to decide if textile waste should be recycled, mended, or used for other purposes. [17]. Defective areas can be found by automated fabric inspection utilizing image-based deep learning. [17]. Defect information affects whether a textile is more suited for recycling, upcycling, repair, or reuse in circular economy contexts. [4, 17].

d. Sustainability and Traceability

Environmental evaluation is essential because textile circularity should not be evaluated just because of economic value. [11, 6]. A foundation for comparing the environmental impacts of recycling, reuse, and other circular behaviors is provided by Life Cycle Assessment studies. [5, 25]. To improve material provenance transparency, Digital Product Passport research encourages the preservation of organized product data and lifetime events. [7, 15].

e. AI Pricing, Recommendation, and Predictive Analytics

Recommendation techniques may evaluate potential users based on item and user attributes [14], while machine-learning regression can estimate continuous market values from structured data [4, 14] Operational planning and marketplace optimization can benefit from predictive analytics [9]. Instead of addressing pricing, matching, and marketplace prediction as separate services, TexCycle AI integrates these capabilities with textile condition and sustainability qualities [4, 6, 9] .

II. RESEARCH GAP AND CONTRIBUTIONS

a. Research Gap

Research Direction	Main Focus	Remaining Limitation / Research Gap
Textile AI Sorting	Material classification	Does not include marketplace functions
NIR/CNN Textile Recognition	Fiber identification	Requires specialized sensing equipment and datasets
Textile Sustainability	Environmental assessment	Often separated from commercial decision-making
Digital Product Passports	Lifecycle traceability	Not necessarily integrated with AI-based marketplace functions
Circular Textile Marketplaces	Buyer–seller connection	Limited integration of AI-based prediction and decision support

Price Prediction	Commercial valuation	Not always combined with garment condition and sustainability factors
Recommendation Systems	Matching users with suitable items	Textile-specific circular-economy context remains limited
Marketplace Prediction	Demand and sales behavior	Rarely integrated with textile sustainability and AI-based material/condition assessment.

Integrating commercial intelligence, traceability, sustainability intelligence, textile intelligence, and predictive marketplace analytics into a single Sri Lankan textile-waste process is the biggest research gap that TexCycle AI fills [4, 5, 7] and [9]. Therefore, rather than asserting that textile markets are wholly novel, the contribution is the integrated and localized decision-support architecture. [4, 12, 6] .

b. Contributions Summary

- C1: Textile intelligence: integrated fabric categorization [12, 13] condition indicators [17] and defect evaluation [17]
- C2: Sustainability intelligence: circular-action decision support [11, 6] and environmental metrics [5, 6]
- C3: Commercial intelligence: buyer-category recommendation [14] and AI-assisted valuation [4]
- C4: Marketplace information, including sales time, listing success, demand, and trend forecasting [9]
- C5: Localization in Sri Lanka: local districts, provinces, currency, industrial kinds, and stakeholder groups [20, 19, 20].
- C6: AI analysis → sustainability → commercial intelligence → marketplace → transaction → traceability – traceable circular process [3, 7, 15], [12].

III. METHODOLOGY AND SYSTEM ARCHITECTURE

a. Research Design

The work combines experimental machine-learning development with a Design Science Research approach [4, 9]. After issue identification and literature review, the research process moves to data gathering, preprocessing, model creation, assessment, API integration, and system-level validation. [12, 17]. While the entire platform is assessed as an integrated decision-support system, each AI component is assessed based on the type of task it performs

[4, 7].

b. Overall Architecture

There are five tiers to the platform:

- Layer 1: User Interface: Marketplace and AI analysis interfaces, as well as dashboards for buyers, sellers, administrators, and super administrators, are provided using TypeScript, React, Vite, and Tailwind CSS.
- Layer 2: Backend API: Users, listings, offers, transactions, notifications, authentication, authorisation, and traceability are all managed using Node.js and Express.js [7].
- Layer 3: AI/ML Services: Trained machine-learning models are hosted using Python and FastAPI [12, 13, 17].
- Layer 4: Data Layer: Users, textile records, AI analysis, sustainability results, predictions, listings, buyers, transactions, and traceability events are all maintained via database-backed storage [7, 8].
- Layer 5: Marketplace Intelligence: Search, suggestions, demand data, price forecasting, listing success forecast, sales time forecast, and trend projection [9].

c. Fabric Intelligence Method

Uploaded textile photos are transformed into structured characteristics by Fabric Intelligence. [12, 13]. A 16-class EfficientNet-B3 image classifier with an ImageNet-pretrained backbone and 224x224 RGB input is used in the present implementation [12, 9]. Blended, Chenille, Corduroy, Cotton, Crepe, Denim, Fleece, Leather, Linen, Nylon, Polyester, Satin, Silk, Terrycloth, Viscose, and Wool are among the project classifications. [12, 13] .

Class filtering, class balancing, AdamW optimization, learning-rate scheduling, and early stopping are all part of the training pipeline [12, 9] . The optimal checkpoint is chosen based on validation accuracy.

A distinct binary classifier based on MobileNetV2 that predicts Defect or No Defect is the implemented defect component [17]. True pixel-level defect segmentation is not carried out by the present backend. [17]. Rather, a display-oriented defect-area indicator is created from defect probability, from which a health score is obtained [17]. The obtained health score and fabric-specific baseline lifetime values are then used to assess repairability and remaining lifespan [4, 17].

Component	Model / Approach	Output
Fabric Classification	EfficientNet-B3, 16 classes	Fabric type and confidence score

Defect Detection	MobileNetV2, binary classification	Defect / No Defect
Health Assessment	Rule-based interpretation	Health score
Lifespan Estimation	Fabric baseline × health ratio	Estimated remaining lifespan

d. Sustainability Intelligence Method

Material and condition characteristics are transformed into environmental and circularity indicators by Sustainability Intelligence. [5, 6]. In addition to fabric type, weight, health score, repairability, condition, district, province, and industry type, the implementation makes use of structured LCA-informed parameters [5, 25] .

Six XGBoost regression estimators are included in the pipeline for:

- Score for circularity [11, 6]
- Scores for recyclability [11, 6]
- % of carbon reduction [5, 25]
- % decrease in water [5, 25]
- CO2 savings [5, 6]
- Water conservation [5, 6]

A different XGBoost grade classifier is then fed the six regression outputs [6]. To ensure consistency in training and inference, different preprocessing processes are employed for the regression and grade-classification stages. [4, 6] . Model outputs and known inputs are used to calculate additional variables, including energy saved, landfill diverted, SDG effect, and economic impact [5, 6] .

Instead of using actual physical measurements, environmental values are estimations based on accepted criteria and assumptions. [5, 25] . Versioned environmental variables, uncertainty ranges, and per-kilogram effect normalization are examples of future improvements [5, 9] .

e. Commercial Intelligence Method

The price-prediction component uses market and textile characteristics to determine the reference price per kilogram [4]. Fabric type, condition, quantity, location, demand, competition, market growth, average market price, seller qualities, and economic indicators are all taken into account in this project [4, 14] .

Buyer categories are ranked by the buyer-recommendation component based on how well textile qualities match customer criteria [9]. Among the categories of candidates are:

- Textile recyclers [4]
- clothing producers [4]
- Designers of fashion [4]
- Furniture producers [4]
- makers of insulation [4]
- producers of cleaning cloths [4]
- Exporters [4]
- Toy producers [4]

Material and event records encompassing analysis, listing, offer, transaction, delivery, and cyclical impact are used to preserve traceability. [7, 8] .

f. Marketplace Predictive Analytics Method

For the purpose of developing prototype models, the marketplace dataset presently has 36 variables and 100,000 records [9]. Fabric type, condition, health score, defect area, repairability, remaining lifespan, weight, sustainability and green scores, CO2 and water savings, anticipated price, buyer type, demand level, time features, district, province, competition indicators, market information, seller experience, listing status, days to sell, and market trend are all significant factors [4, 9] .

This dataset is intended to illustrate the technological workflow rather than to assert real-market reality because it is synthetic. [9, 10] .

Model	Target	Algorithm
Listing Success	Sold / Unsold	XGBoost Classifier
Sales Time	Days to Sell	XGBoost Regressor
Demand Level	Low / Medium / High	Classification Model
Market Trend	Increasing / Stable / Decreasing	XGBoost Classifier
Buyer Recommendation	Buyer Compatibility	Random Forest + XGBoost

g. Data Preprocessing

Before inference or training, uploaded photos are normalized and scaled for image-based models [12, 17] . Because illumination, backdrop complexity, and viewing distance might impact prediction confidence, the method takes image quality into account. [12, 17]. Numerical variables are verified and transformed into numerical form for tabular models, and learned encoding pipelines are used to handle categorical data [4, 9]. Data cleaning, feature engineering,

exploratory analysis, and separating training and assessment data are all part of dataset preparation [4, 9] .

h. Model Training and Evaluation

The choice of model depends on the job. [12, 17] . While gradient-boosted tree techniques are employed for structured tabular prediction [4, 9], deep learning architectures are utilized for image classification and defect identification [12, 17]. The following is the training workflow:

Data gathering → cleaning → exploratory analysis → feature engineering → train/test split → preprocessing → baseline model → model training → hyperparameter tuning → evaluation → best model selection → model saving → API integration → application testing.

Component	Problem Type	Current Model / Approach	Main Evaluation Metrics
Fabric Classification	Multiclass Classification	EfficientNet-B3	Accuracy, Precision, Recall, F1-Score
Defect Detection	Binary Classification	MobileNetV2	Accuracy, Precision, Recall, F1-Score
Sustainability	Multi-output Regression	XGBoost Estimators	MAE, RMSE, R^2
Price Prediction	Regression	XGBoost Regressor	MAE, RMSE, R^2
Demand Prediction	3-Class Classification	Trained ML Classifier	Accuracy, Precision, Recall, F1-Score
Market Trend	3-Class Classification	Trained Model + Label Encoder	Accuracy, Precision, Recall, F1-Score
Listing Success	Binary Classification	Trained Classifier	Accuracy, Precision, Recall, F1-Score
Sales Time	Regression	XGBoost Regressor	MAE, RMSE, R^2

i. System Integration and Deployment

The Node.js/Express backend handles authentication, authorization, users, listings, offers, transactions, alerts, and traceability [7]. Python/FastAPI services host the trained machine-learning models. [12, 17] . Database-backed storage keeps track of marketplace and lifecycle records, while structured environmental factors can be maintained independently [7, 8] . This separation enables AI services to be updated independently of the web application [4, 7] . The presentation layer makes use of React, Vite, TypeScript, Tailwind CSS, React Router, and Lucide icons. [4].

IV. SYSTEM WORKFLOW AND IMPLEMENTATION

a. End-to-End Workflow

The workflow for operations is:

Image upload, image preprocessing, fabric categorization and flaw analysis, condition evaluation, sustainability assessment, price prediction, buyer suggestion, marketplace prediction, listing, offer, transaction, delivery, and impact record [4, 7, 9] .

Unstructured textile waste is transformed by this process into structured, AI-assessed material, which is then turned into sustainable commercial inventory [4, 6, 9].

b. Seller workflow

- Run Fabric Intelligence after uploading pictures of textile waste [12].
- Examine fabric type, confidence, information about defects, outputs connected to condition, and remaining lifespan [12, 17] .
- View environmental indicators and sustainability grades [5, 6].
- View the anticipated sales time, buyer suggestions, trend, listing success, demand, and price [4, 14, 9] and [9].
- Make and maintain marketplace listings, as well as monitor orders and offers [7] .

c. Buyer workflow

- Look up and sort the textile materials that are accessible [14].
- Examine market data, sustainability, material, and condition [4, 6].
- Examine vendor details and make purchases or offers [7].
- Keep track of orders and finish the rating and review processes [7].

d. Re - analysis and Re - grading

The implementation supports re-running analysis on existing listings [4]. Re-grading recalculates sustainability outputs using stored AI analysis, while re-analysis reruns fabric/defect analysis using stored listing images and then re-grades sustainability [4, 6]. This is important because existing listings should not remain permanently tied to an older model version when model logic is improved [4, 6].

e. Traceability

Alongside textile, AI, sustainability, price prediction, buyer recommendation, listing, offer, transaction, and market snapshot data, a Traceability Event entity is kept up to date [7, 8]. One way to depict the lifecycle is:

- Analysis → Listing → Offer → Transaction → Delivery → Impact → Upload [7, 15].

This maintains a digitized history of significant textile-related events and promotes openness. [7, 8].

V. RESULTS AND EVALUATION

a. Fabric Intelligence Results

89.68% test accuracy for 16-class fabric classification was attained in the most recent EfficientNet-B3 experiment [12, 18]. Prior to an early termination, the training procedure achieved a best validation accuracy of around 91.41% [12, 18] . 90.87% was the stated defect-detection result [17].

Model Evaluation Metric	Result
Fabric Type Classification	89.68% Accuracy
Defect Detection	90.87% Accuracy

While class-level accuracy, recall, and F1-score are still crucial for a thorough publication-grade review, these numbers show that the image-based technique is technically promising for automated preliminary assessment [12, 17, 18] and [18].

b. Sustainability Intelligence Results

For every stated aim, the sustainability trials yielded R^2 values greater than 0.93, with the maximum R^2 of 0.9887 for CO2-saving prediction and 0.9854 for water-saving prediction [5, 7].

Model	MAE	RMSE	R^2
Circularity Score Prediction	2.429	3.043	0.9657
Recyclability Score Prediction	2.459	3.088	0.9502
Carbon Reduction Prediction	2.397	3.009	0.9589
Water Reduction Prediction	2.464	3.084	0.9389
CO_2 Saving Prediction	4.456	6.110	0.9887
Water Saving Prediction	310.365	437.567	0.9854

The findings imply that the correlations shown in the development data may be modelled by the used XGBoost regression pipeline. [5, 6]. However, high R2 on generated or derived data should not be taken as evidence that the estimates correspond to actual facility environmental observations [5, 9].

c. Marketplace Intelligence Results

Model	Evaluation Metric	Result
Price Prediction	R^2	0.9590
Price Prediction	MAE	12.24
Price Prediction	RMSE	15.40
Sales-Time Prediction	R^2	0.7610
Demand Prediction	Accuracy	99.54%
Listing Success Prediction	Accuracy	83.65%
Market Trend Classification	Accuracy	98.72%
Buyer Recommendation/Matching	Accuracy	83.56%

The probability distributions displayed in the program are subject to a crucial interpretation rule. [9]. A number like 99.72% for one trend class does not indicate the validation accuracy of the training model; rather, it indicates the confidence/distribution of a single prediction [9]. Therefore, it is necessary to present model-level accuracy, precision, recall, F1-score, MAE, RMSE, and R^2 independently from individual prediction outputs [9, 18].

d. Integrated Demonstration

Textile photos are typically the first step in a seller's process. [12]. Information on fabric type and condition is generated by Fabric Intelligence [12, 17] . These characteristics are transformed into environmental indicators and a sustainability grade by Sustainability Intelligence. [5, 6] . Next, a reference price and buyer recommendations are provided by commercial intelligence [4, 14] . Demand, listing success, sales time, and trend data are added by Marketplace Predictive Analytics. [9]. After that, the structured record may be connected to offers, transactions, delivery events, and circular-impact data and published as a marketplace listing [7, 15].

The present interface shows the following for a 50-kilogram example Denim Textile Waste listing:

- Price prediction: LKR 260.15/kg [4].
- LKR 109.16/kg is the listed price [4].
- LKR 13,007.50 is the estimated total value [4].
- Level of demand: high [9].
- Textile recyclers are the best buyer match [14].

- Market trend: Growing with a 70% confidence level [9].
- Probability of listing success: 74% [9].
- Sales time estimate: around eighteen days [16].

These should not be used as independent market measurements because they are interface outputs for a prototype listing [9, 10] .

VI. DISCUSSION

a. Integrated Intelligence Value

The combination of several specialized intelligence layers rather than a single high-performing model is the primary technological benefit of TexCycle AI [4, 7, 9] .

- What is this material and what state is it in? Fabric Intelligence responds [12, 17] .
- What is the environmental importance of Sustainability Intelligence? [5, 6]
- Commercial intelligence responds: How much is it worth, and who may purchase it? [4, 14]
- Marketplace Predictive Analytics responds to the question, "How is it likely to perform in the marketplace?" [9]

Unstructured textile waste is transformed by the architecture into structured AI-assessed material, sustainable commercial inventory, and a traceable circular transaction. [4, 7]. Price alone can influence decisions that disregard environmental value, which makes this integration important [6, 25]. If reuse prevents significant resource usage, a less expensive batch may nonetheless have a high circular value [11, 5].

b. Model-Level Interpretation

On its assessment dataset, the price-prediction experiment produced a high R2 of 0.9590 with comparatively low MAE and RMSE [4] .However, the outcome mainly shows that the selected feature set and model pipeline can learn the created association because the dataset is synthetic [4, 9] . It shouldn't be used as proof that the model would perform similarly on actual transactions in Sri Lanka [9, 10] .

A deployment trade-off is demonstrated by the Fabric Intelligence architecture [12]. For server-side inference, EfficientNet-B3 offers robust visual representation while maintaining practicality [12, 18]. The independent MobileNetV2 fault classifier is lightweight and well-suited for a responsive API process [17]. The present health and defect-area outputs should be referred to as derived indicators rather than direct measurements of damaged pixels, since the defect model is a binary classifier rather than a segmentation network [17].

Because each environmental objective has its own estimator, the sustainability pipeline benefits from the employment of several XGBoost regressors [5, 6] Predicted sustainability qualities are transformed into a grade by a second XGBoost classifier [6]. By mirroring the data available at inference time, this staged architecture prevents the classifier from receiving ground-truth environmental outputs when such values would not be accessible during deployment [5, 6].

c. Practical Value

Sellers and buyers may find it less necessary to transition between different tools thanks to the integrated process [4, 7]. Before publishing a listing, a seller might transition from image-based material assessment to sustainability data, market valuation, buyer discovery, and marketplace forecast [4, 12, 9]. Structured textile qualities and sustainability data can enhance material comparison and discovery for consumers [14, 6].

VII. LIMITATIONS AND FUTURE WORK

a. Limitations

1. Synthetic marketplace dataset: Model performance does not demonstrate actual Sri Lankan market accuracy because the existing marketplace forecast dataset is synthetic [9, 6].
2. Limited actual transaction data: A marketplace cold-start problem is caused by the scarcity of genuine previous pricing and transaction histories [4, 9].
3. Image limitations: RGB images are not always able to accurately identify the precise fibre composition or chemical contamination [13, 26, 27]
4. Environmental estimation: LCA elements, assumptions, and environmental circumstances affect environmental outputs, including CO2 and water savings [5 [25]].
5. Model drift: Over time, changes in textile pricing and demand may result in model-drift risk [9].
6. Validation of buyer recommendations: In the end, genuine user interactions, offers, or completed transactions must be used to validate buyer-recommendation performance [14].

b. Future Work

1. Gather actual transaction histories and bigger, verified Sri Lankan textile picture databases [4, 12] and [9].
2. Combine dynamic pricing with real-time market data [4, 9] .
3. Create sophisticated buyer recommendations, clever logistics, and reverse auctions [14].
4. Verify environmental reporting and extend digital product passports [7, 8].

5. Describe model-drift monitoring and AI-based fraud detection [9].
6. Offer assistance in Sinhala, Tamil, and English [19].
7. Retrain and update models regularly in response to fresh, verified data [4, 9].
8. Extend validation to further textile markets in South Asia [1].
9. Enhance defect analysis by using proven pixel-level segmentation instead of binary classification [17].
10. Add contaminant identification, multi-image analysis, and human-reviewed confidence thresholds [12, 17].
11. For environmental reporting, use versioned LCA factors, uncertainty ranges, and per-kilogram normalization [5, 10].

VIII.CONCLUSION

This study introduced TexCycle AI, an AI-driven circular economy intelligence platform and textile-waste marketplace created for Sri Lanka [4, 7]. Four research components are integrated into a continuous workflow that covers material assessment, environmental valuation, pricing, buyer matching, marketplace prediction, and traceability [7, 14]. These components are Fabric Intelligence. [12, 17], Sustainability Intelligence [5, 6], Commercial Intelligence [4, 14], and Marketplace Predictive Analytics [9].

Technical viability is demonstrated by the current prototype. While the claimed defect-detection result was 90.87%, fabric-type categorization reached 89.68% test accuracy [12, 17, 18]. R2 values for sustainability trials ranged from 0.9389 to 0.9887 for all stated aims [5, 6]. The price-prediction experiment yielded R^2 0.9590, RMSE 15.40, and MAE 12.24 [4]. 99.54% demand accuracy, 83.65% listing-success accuracy, 98.72% market-trend accuracy, and 83.56% buyer-matching accuracy were attained in marketplace trials [14, 9].

Because the marketplace dataset is artificial and environmental effects rely on accepted LCA assumptions, these findings should be viewed as prototype evidence [5, 9, 10]. The integration and localization of AI-enabled textile circularity and marketplace intelligence in Sri Lanka is the main contribution [4, 7, 19]. The next crucial step toward evidence-based adoption is real-world validation with purchasers, recyclers, textile producers, and other industry **players [1, 20, 9].**

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